FinBERT Quantum Optimization — Summary Report

Setup

Base checkpoint: ahmedrachid/FinancialBERT (pre-trained on financial corpora, no sentiment fine-tuning). ProsusAI/finbert was excluded because it is already fine-tuned on Financial PhraseBank, inflating results to 97%+ regardless of hyperparameters. Two dataset tracks were maintained separately throughout: the primary CSV (4,846 sentences, split 3,392/727/727) and the Financial PhraseBank AllAgree subset (2,264 sentences, split 1,584/340/340). Both splits are 70/15/15 stratified with seed=42. Label mapping: negative=0, neutral=1, positive=2.

Note on baseline values: An earlier progress update (July 1) reported 84.3% accuracy on the primary track and 97.1% on the AllAgree track. The final results reported here (82.4% and 97.6%) differ because all baselines were re-run after correcting a dropout implementation bug, ensuring the baseline, QUBO-optimized, and sweep-best models were all trained under identical conditions. The earlier results used a version of the code that did not apply the dropout parameter to the model's internal layers.

Process

All 32 hyperparameter combinations from the paper's 5-bit QUBO encoding (4 learning rates × 2 dropout rates × 2 batch sizes × 2 epoch counts) were trained and evaluated on the validation set. The 32 validation losses were used to construct a 5×5 QUBO matrix via least-squares regression with baseline subtraction. Three optimizers — SA (10 runs, 500 iterations, geometric cooling), QAOA (Qiskit StatevectorSampler, COBYLA, depths p=1,2,3), and VQE (SamplingVQE, RealAmplitudes + EfficientSU2 ansatze, COBYLA + Nelder-Mead) — were run on local noiseless simulation to find the QUBO minimum. All three converged to the same bitstring on both tracks. Final models were trained from scratch with the baseline, QUBO-optimized, and sweep-best (actual lowest validation loss) configurations and evaluated on the held-out test set.

Results

Primary CSV Track			
Method	**Hyperparameters**	**Accuracy**	**Macro F1**
Baseline	LR=2e-5, DO=0.1, BS=32, EP=3	82.4%	0.81
SA/QAOA/VQE	LR=1e-5, DO=0.3, BS=16, EP=3	80.9%	0.80
Sweep Best	LR=5e-5, DO=0.3, BS=32, EP=3	82.9%	0.82

AllAgree Track			
Method	**Hyperparameters**	**Accuracy**	**Macro F1**
Baseline	LR=2e-5, DO=0.1, BS=32, EP=3	97.6%	0.97
SA/QAOA/VQE	LR=2e-5, DO=0.1, BS=16, EP=4	97.1%	0.96
Sweep Best	LR=5e-5, DO=0.1, BS=16, EP=4	96.2%	0.95

All three optimizers converged to the same solution on both tracks (SA: 10/10 runs, QAOA: all depths, VQE: all 4 ansatz/optimizer combinations). This is consistent with the reference paper and expected for a 5-qubit problem. The QUBO-optimized hyperparameters did not outperform the baseline on either track's test set. Repeated trials (5 runs per configuration with different random seeds) confirmed this: on the primary track, baseline accuracy was 83.2% ± 0.9% and sweep-best was 83.6% ± 0.4% (95% CIs computed as mean ± 1.96 × std/√5 across 5 random seeds for model initialization) , with overlapping confidence intervals. The QUBO-optimized configuration consistently underperformed at 81.9% ± 0.7%. On the AllAgree track, all three configurations fell within 96–98% with overlapping intervals.

Reflection

Throughout the process, there were several issues encountered. Initially, there was a bit of floundering attempting to identify a base checkpoint that had no sentiment fine-tuning.

In implementation, another error was found as the dropout config value was passed through the code but never applied to the model's internal layers (hidden_dropout_prob, attention_probs_dropout_prob, classifier_dropout). This was discovered when dropout=0.3 configs produced identical losses to dropout=0.1. After the fix, the full 32-combination sweep was re-run.

Finally, the results did not necessarily match the original paper, as the results underperformed. There are several primary reasons for this:

1. Checkpoint: Examination of the provided reference model files revealed the original study used yiyanghkust/finbert-tone, a BERT model already fine-tuned for financial sentiment classification. This study used ahmedrachid/FinancialBERT, a BERT model pre-trained on financial corpora but with no sentiment fine-tuning, as required by the instruction document's directive to build the pipeline independently from raw data. A pre-fine-tuned checkpoint may exhibit greater sensitivity to hyperparameter variation because its classification performance starts lower, creating more room for optimization to demonstrate improvement. The ahmedrachid/FinancialBERT checkpoint used here achieved strong baseline performance (82.4% primary, 97.6% AllAgree), leaving less headroom for tuning gains.
2. AllAgree ceiling: The 100% agreement subset is inherently unambiguous — any BERT-class model achieves 95%+. The paper's reported 85% baseline is anomalously low, and their confusion matrix implies ~968 test samples versus our 340 from a standard 15% split, suggesting a different data construction.
3. QUBO approximation: The QUBO identified the minimum, as intended, but it did not correspond to the best configuration. The quadratic surrogate (15 parameters fit to 32 noisy validation losses) cannot capture higher-order interactions between hyperparameters, a limitation to QUBO approximation.

QUBO surrogate diagnostics confirm this limitation: the quadratic model achieved R^2=0.67 (primary) and R^2=0.69 (AllAgree), explaining roughly two-thirds of the variance in validation loss. Spearman rank correlation was 0.85 and 0.88 respectively ($p<0.001$), indicating the approximation captures the general ranking of configurations but misranks top candidates. The QUBO's predicted best did not match the actual best on either track, with top-5 overlap of only 2/5 (primary) and 3/5 (AllAgree).

Reproducibility

All code is in src/ and can rebuild the pipeline end to end: prepare_data.py → build_qubo.py → run_sa.py → run_qaoa.py → run_vqe.py → evaluate_optimized.py. All metrics are saved as JSON in outputs/metrics/. Random seed 42 is applied to data splits, PyTorch, and NumPy. Quantum runs used local simulation only — no IBM Quantum hardware was consumed. Full deliverables with confusion matrices, loss curves, and detailed comparison notes are provided in the accompanying report.

The complete codebase, processed data, metrics, and reports are available at https://github.com/fzara007/finbert-quantum-optimization